Tahmid Hasan Sakib

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PROFESSIONAL SUMMARY

Graduate researcher evaluating adversarial traffic-sign perception, runtime anomaly monitoring, image restoration, and independent intrusion detection. Connect detector evidence to physical vehicle outcomes with Python and embedded AI platforms.

TECHNICAL SKILLS

Machine learning and software: Python, PyTorch, scikit-learn, TensorFlow/Keras, NumPy/Pandas, OpenCV, Git, Linux.

Perception and edge inference: YOLOv5n/YOLOv8, NVIDIA Jetson TX2 and AGX Orin, study-specific TensorRT, CSI/RealSense cameras, CARLA.

EXPERIENCE

Graduate Research Assistant | Tennessee Technological University, Cookeville, TN | May 2024-Present

- Evaluated YOLOv5n stop-sign detection and camera/LiDAR evidence under printed patches in physical QCar1 runs.
- Compared clean-calibrated PULSE detector evidence in recorded-log replay and distinct live QCar2 fallback tests.
- Analyzed REDAP attacked/restored QCar2 frames offline and documented separate later closed-loop outcomes with collaborators.
- Mentored undergraduate researchers in NSF REU cohorts in summers 2025 and 2026 under faculty supervision.

Hardware Engineer Intern | Robi Axiata Limited, Dhaka, Bangladesh | December 2019-January 2020

- Prototyped AVR/Arduino systems and simulated hardware and communication behavior with MATLAB, LTspice, and Cisco Packet Tracer.

SELECTED AI AND COMPUTER VISION PROJECTS

QCar1 physical adversarial perception | YOLOv5n and sensor fusion

Runs with a missed stop changed from 8/18 to 0/18 across separate pre/post physical groups; detector and runtime settings both changed.

PULSE runtime anomaly monitoring | physical QCar2

Clean-calibrated monitoring flagged 18/20 patched-sign encounters in 10 live fallback-on runs; all 20 stop sequences completed, without an independent stop-line measure.

REDAP restoration-based defense | offline physical frames

At a 0.60 detector-confidence cutoff, 57/151 attacked and 110/151 restored QCar2 frames qualified; later vehicle tests were separate.

Network Intrusion Detection for IoT Devices | GitHub repository

Independent post-undergraduate CICIoT2023 notebooks cover binary, 8-class, and 34-class tasks, feature selection, conventional ML, and ANN/LSTM/GRU.

Visual Product Recommendation System Using CNN

Earlier Python project exploring image-based recommendation of visually similar retail products.

EDUCATION

Tennessee Technological University | M.S., Electrical and Computer Engineering, expected December 2026 | GPA 4.00/4.00

Master's thesis defended; official degree conferral pending.

Military Institute of Science and Technology, Dhaka, Bangladesh | B.Sc., Computer Science and Engineering, August 2021

SELECTED RESEARCH OUTPUTS

First author, IEEE Access adversarial-perception article, accepted for publication, 2026; coauthor, physical naturalistic-patch paper, IEEE SATC 2026; first author, tweet-classification paper, IEEE ACMI 2021.